

# Adam Zhu

(515) 708-7879 | [adamzhu@andrew.cmu.edu](mailto:adamzhu@andrew.cmu.edu) | [linkedin.com/in/adam-zhu-cmu](https://www.linkedin.com/in/adam-zhu-cmu) | [github.com/adam-zhu1](https://github.com/adam-zhu1) | [adam-zhu1.github.io](https://adam-zhu1.github.io)

## Education

### Carnegie Mellon University

Pittsburgh, PA

*Bachelor of Science in Statistics and Machine Learning; Minor in Computer Science*

## Experience

### SURF Research Intern, Research Data and Computing Office

May 2026 – Aug. 2026

*National Institute of Standards and Technology (NIST)*

*Gaithersburg, MD*

- Built a reproducible 5-stage Python pipeline that separates bot from human traffic in NIST's public data portal logs; per-IP behavioral features and 4 scikit-learn classifiers reached  $F1 = 0.95$  and showed that 51% of 190,687 requests were automated.
- Designed a second-pass behavioral detection rule that caught a disguised scraper: 50,000+ spoofed requests had passed as human, and correcting their labels improved all 4 classifiers.
- Found that 23% of 20,790 human searches returned no results by analyzing bot-filtered search behavior, then turned this unmet-demand finding into recommendations for NIST's search and dataset-acquisition priorities.

### Research Intern

Jun. 2023 – Aug. 2024

*Center for Statistics and Applications in Forensic Evidence, Iowa State University*

*Ames, IA*

- Co-authored a manuscript, now under review at *Forensic Sciences*, that evaluates camera fingerprinting for device identification across 24,000+ camera images.
- Developed document-degradation methods and produced a dataset of 100+ degraded writing samples so that handwriting-similarity analysis could be tested under realistic conditions.

### Research Intern

Jan. 2024 – Jun. 2024

*Virtual Reality Applications Center, Iowa State University*

*Ames, IA*

- Fit regression models of time-to-escape across 454 simulated VR evacuation trials with 227 participants; smart audio guidance cut evacuation time by a statistically significant margin over static and adaptive signage. Co-author of a manuscript in preparation.

## Projects

### TrueLine: iOS Bowling Ball Tracker | Swift, SwiftUI, Core ML, Python, PyTorch, OpenCV

2025 – Present

- Developing an iOS app that turns a single iPhone into a ball-motion tracker for per-throw launch speed, breakpoint, and entry angle. These metrics otherwise require a \$10k+ in-lane installation; App Store launch is in preparation.
- Fine-tuned a YOLOv8 detector, starting from hand-labeled seed frames and growing the training set with a model-assisted labeling loop, then exported it to Core ML with Kalman-filter tracking and Savitzky-Golay smoothing. The full computer vision pipeline runs on-device with zero third-party dependencies.
- Added support for arbitrary camera angles: video maps into real-world lane coordinates through a homography from a 4-corner calibration, where a custom gutter-line detector proposes the corners and a magnifier-loupe drag UI refines them.
- Wrote a clip-by-clip parity harness that checks the Swift port against the Python/OpenCV prototype; board position agrees within 1 board and launch speed within 1–2%.

## Publication

Lin, H., **Zhu, A.** and Wang, C. (2024). Statistical Tests for Proportion Difference in One-to-Two Matched Binary Diagnostic Data: Application to Environmental Testing of Salmonella in the United States. *Mathematics* 2024, 12, 741. [doi.org/10.3390/math12050741](https://doi.org/10.3390/math12050741)

## Leadership

### Co-Captain

Jun. 2021 – Jun. 2025

*Team Neutrino, FIRST Robotics Team #3928*

*Story County, IA*

- Managed the 27-sponsor corporate and community network that sustained the team's \$32,000 annual budget, and led 7 sub-teams with 40+ members. The team earned multiple NASA Engineering Inspiration Awards and World Championship qualifications over this span.

### Camp Co-Designer

Summers 2024 & 2025

*COR Robotics Summer Camps*

*Central Iowa*

- Co-designed a project-based robotics curriculum, from drone racing to battle drones, and taught foundational STEM concepts with it to 100+ students in grades 3–8.

## Technical Skills

**Languages:** Python, Swift, R, Java, C, Bash, MATLAB,  $\LaTeX$

**Frameworks & Libraries:** pandas, scikit-learn, PyTorch, OpenCV, FastAPI, SwiftUI, Core ML, AVFoundation

**Tools & Data:** Git/GitHub, Jupyter, matplotlib, MongoDB, REST APIs, JSON/JSON-LD, regex

**Statistics & ML:** Classification, Feature Engineering, Model Evaluation, Experimental Design, GLMs, ANOVA

## Honors & Awards

Congressional Award (Silver Certificate), The Congressional Award Foundation, U.S. Congress

Jan. 2025

1st Place (Mathematics Category) & RAYGUN Innovative Scientist Award, SSTFI

Apr. 2024

Eagle Scout Rank, Boy Scouts of America

May 2023

World Championship Engineering Inspiration Award, FIRST Robotics (1 of 6 in the world)

Apr. 2022